Zoe Bishop

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EDUCATION

Cornell University; Ithaca, NY

Master of Engineering, Mechanical Engineering — Expected May 2027

Bachelor of Science, Mechanical Engineering | GPA: 3.5 | Dean's Honor List five semesters | Cum Laude — May 2026

Relevant Courses: Systems Engineering and Six Sigma for the Design and Operation of Reliable Systems, Finite Element Analysis for Mechanical and Aerospace Design, Dynamics of Flight Vehicles, Propulsion of Aircraft & Rockets, Project Management

PROFESSIONAL EXPERIENCE

BETA Technologies - *Product Quality Engineering Intern*; South Burlington, VT & Remote — Jun 2026 - Present (PT)

- Led an environmental continuous improvement project for battery manufacturing and storage. Conducted corrective and preventive actions (CAPA) and root cause analysis (RCA) within an AS9100 QMS. Developed and executed a test plan, analyzed data, and presented findings that led to a design change. Reported to the biweekly Corrective Action Board (CAB).
- Supported FAA conformity documentation and traceability for the all-electric aircraft program.
- Created a dashboard for tracking nonconformance reports (NCRs) and corrective actions for top quality issues.
- Wrote four training guides for battery manufacturing line tools.

GlobalFoundries - *Metrology Equipment Engineering Intern*; Essex Junction, VT — Jun 2025 - Jan 2026

- Led the first-of-a-kind qualification of a measurement toolset through the Product Change Review Board (PCRB).
- Evaluated wafers to identify sample sets and performed gage repeatability and reproducibility (GR&R) studies for measurement system analysis (MSA). Analyzed and assessed risks of the toolset using a change risk assessment; presented findings to the Failure Mode and Effects Analysis (FMEA) Review Board.

Seattle University - *Environmental Engineering Intern*; Seattle, WA — Apr 2022 - Aug 2023

- Implemented and programmed sensor instrumentation (temperature, pH, dissolved oxygen, electrical conductivity, flow, and pressure transducer) for a hydroponics system using Arduino and Raspberry Pi; created video and written operating manual.

PROJECTS

Sagitta Borealis Systems - *Additively Manufactured Spacecraft Thruster M.Eng. Project*; Ithaca, NY — Sep 2026 - Present

- 3D printing porous electrospray emitter arrays (350–500+ emitters) on a Nanoscribe GT2 for VLEO satellite propulsion.
- Measuring print variability with laser profilometry, testing fluid flow, and documenting process iterations.

Engineers in Action, Cornell Project Team - *Quality Control Engineer*; Ithaca, NY & Eswatini, Africa — Sep 2023 - May 2025

- Collaborated with a rural community of ~5,400 people and 10 travel team members to build an 88-meter suspended pedestrian bridge to enable safer and faster passage over a river during dangerous conditions.
- Monitored quality control during construction. Audited and recorded as-built plans.
- Documented design and implementation plans and presented in review calls with licensed engineers.

Cornell University - *2025 Mechatronics Cube Craze Competition*; Ithaca, NY — May 2025

- Designed and coded an Arduino-based robot in a team of three to collect the most blocks in an arena; placed 2nd of 45 teams.

LEADERSHIP EXPERIENCE/ACTIVITIES

Cornell University - *Cornell Club Soccer and Cornell Wild Roses Ultimate Frisbee*; Ithaca, NY — Aug 2022 - Present

- 2025 President of Cornell Club Soccer; 2025 region MVP; 2022, 2023, 2025 NIRSA National Tournament qualifier.
- Social media manager of the Wild Roses Ultimate Frisbee team; 2025 USA Ultimate D-1 College Championships qualifier.

Cornell University - *Society of Women Engineers (SWE)*; Ithaca, NY — Aug 2022 - May 2026

- *2025-26 Co-President;* oversaw eight directorships and managed planning and logistics for a 200-member organization. Facilitated monthly executive board meetings.
- *2024-25 Co-Outreach Director, 2023-24 DiverSWEty Outreach Committee Lead;* led outreach to support women in STEM.

TECHNICAL SKILLS

Quality - Six Sigma, GR&R, FMEA, RCA, CAPA, AS9100 QMS, NCR, Plex, Microsoft Excel, Grafana

Manufacturing - Cornell Emerson Machine Shop trained (manual mill and lathe), engineering drawing interpretation

Design - Autodesk Fusion 360, Autodesk AutoCAD, Finite Element Analysis (ANSYS), 3DEXPERIENCE

Programming - MATLAB, Python, Arduino IDE, C++, Raspberry Pi, Data Analysis